

TASK-EFFICIENT AGENT EVALUATION USING CACHED TRACES

Anonymous authors

Paper under double-blind review

ABSTRACT

Efficient benchmarking estimates a system’s full-benchmark performance from a small number of task executions. Standard approaches rely solely on the binary correctness of executions from a collection of already-evaluated reference systems. An agentic system, however, produces a rich trace for each execution that can reveal substantially more about its capabilities. Exploiting this information requires useful trace representations. We propose representing a trace by extracting a structured, rubric-guided summary of it with an LLM and embedding the summary with a standard text encoder. These representations preserve correctness- and task-related signal while minimizing the presence of system identity-, model family-, and harness-related signals that dominate raw trace embeddings. Given this representation for each task execution from all reference systems, we construct low-dimensional representations of the systems, predict a new system’s score from its position among the reference systems, and combine the resulting geometric prediction with a two-parameter logistic item-response model. On SWE-bench Verified and Terminal-Bench 2.0, the proposed estimator reduces mean absolute error from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$), respectively, when provided a single random executed task relative to an item-response model alone. These results indicate that trace geometry and correctness outcomes carry complementary predictive signal and enable more accurate benchmark score estimation from limited agent executions.

1 INTRODUCTION

The dominant paradigm of generative model use is increasingly agentic: systems that plan, invoke tools, observe intermediate results, and iterate over long horizons rather than emit a single response (Yang et al., 2024; Wang et al., 2025). As agentic capabilities increase, so does the range of settings in which a given system might be deployed – and, with it, the number of benchmarks on which the system must be evaluated before it can be trusted with consequential work (Kapoor et al., 2026). Agentic evaluation, however, is expensive: each task execution requires a properly sandboxed environment, the tasks themselves are complex and long-horizon, and a single execution routinely consumes hundreds of thousands of tokens (Jimenez et al., 2024; Terminal-Bench Team, 2026). At reported per-task costs of \$0.13–\$4 (an aggregate \$1.84 per rollout across 21,730 rollouts in Kapoor et al., 2026), fully evaluating a single system costs roughly \$100–\$2,000 per benchmark, and the sensitivity of agentic systems to prompts, scaffolds, and underlying model versions means there is always another variant to evaluate (Kapoor et al., 2024). Notably, executing an agentic task produces more than a grade: file diffs, tool invocations, intermediate plans, and error messages are all created along the way. These execution artifacts carry information about a system’s capabilities well beyond the binary “correct” or “not correct” that standard evaluation pipelines retain.

Given the cost of a single full evaluation, efficient evaluation is necessary for increasing the evaluation coverage of a system. Task-efficient (or query-efficient) evaluation – benchmark compression and anchor-item selection (Vivek et al., 2024; Maia Polo et al., 2024), item response theory and adaptive testing (Kipnis et al., 2025; Truong et al., 2025) – reduces the number of required executions by exploiting structure in the item-level scores of previously-evaluated reference systems. These methods typically use only the binary correctness of each execution. A complementary family of methods leverages additional item-level information – such as cached responses or traces – to further improve the modeling of system behavior: responses are embedded with an off-the-shelf em-

bedding function, systems are represented as points in the resulting low-dimensional geometry, and benchmark prediction is cast as inference in that space (Duderstadt et al., 2023; Helm et al., 2025; Zhuang et al., 2025; Helm et al., 2026). Critically, these methods rely on the embedding function to capture what it means for two artifacts to be *similar*. For short natural-language responses, off-the-shelf semantic encoders fill this role well (Reimers & Gurevych, 2019). Agentic traces are different: they are long, semi-structured logs whose surface form is dominated by harness formatting, tool-call syntax, and model-family idiosyncrasies (Sun et al., 2025). Indeed, we find that linear probes recover the identity of the system that generated a raw trace embedding with balanced accuracy up to 0.89 – while representations of what the agent actually did remain comparatively impoverished – making off-the-shelf embedding functions poorly suited for extending geometry-based efficient evaluation to the agentic setting.

Contributions. We make two primary contributions.

1. **Methodological:** We introduce TRUBRIC (*trace rubric*), a representation that extracts a structured, rubric-guided summary of each trace with an LLM and embeds the summary with a standard text encoder. Across seven embedding functions, linear probes show that TRUBRIC representations preserve correctness- and task-related signal while collapsing system-identity decodability from 0.89 to 0.20 balanced accuracy – selective attenuation of confounders rather than erasure of content.
2. **Empirical:** We demonstrate that TRUBRIC-based estimators are query-efficient on SWE-bench Verified (107 systems) and Terminal-Bench 2.0 (65 systems) under a conservative leave-one-model-family-out protocol. Blended with an item-response model, TRUBRIC geometry reduces single-task MAE from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$) on the two benchmarks respectively, correctly orders 88% of system pairs versus 79% for scores alone, and is never less accurate than its score-only counterpart at any budget. Ablations show the gains grow with the reference library, compose with adaptive task selection, and are insensitive to the choice of embedding function.

1.1 PROBLEM STATEMENT

Given an agentic system f , we consider the problem of predicting its score on a benchmark $Q^* = \{q_1, \dots, q_M\}$. For our purposes, a system $f \in \mathcal{F}$ is a random mapping from a query space $\mathcal{Q}$ to a trace space $\mathcal{X}$: executing f on query q yields a trace $f(q)$, a long, semi-structured log of the system’s actions, observations, and outputs. Let $s : \mathcal{X} \rightarrow \{0, 1\}$ denote the harness’s trace-level scoring function and $y : \mathcal{F} \times 2^{Q^*} \rightarrow [0, 1]$ denote the benchmark scoring function, $y(f, Q) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{E}[s(f(q))]$. We refer to $y(f, Q^*)$ as y when the context is clear.

As described above, full evaluation of f on all queries may be infeasible. Instead, we assume access to previously-evaluated (system, score) pairs $(f_1, y_1), \dots, (f_n, y_n)$, along with each reference system’s trace and trace-level score for every query. Such reference caches exist publicly at scale.¹ Given this additional information, our goal is to estimate y with $m \ll M$ queries. That is, we seek to identify an estimate $\hat{y}$ that predicts the full benchmark score of f from its traces (and, optionally, its trace-level scores) on a subset $Q \in 2^{Q^*}$ and the information available from the previously-evaluated systems.

Two aspects distinguish this setting from the short-response setting of Helm et al. (2026). First, the observation associated with each query is not a short response but a trace of 10^4 – 10^6 tokens in one of many mutually incompatible harness formats. Whether and how traces can be embedded usefully is the central question of this paper. Second, the reference population is confounded: reference systems that share the target’s underlying language model or harness permit trivially accurate prediction via score-copying. We therefore evaluate under a leave-one-model-family-out protocol – references sharing the target’s underlying model family are excluded – and verify robustness under a leave-one-harness-out protocol. The query subset Q may be fixed, drawn at random, or selected adaptively per target; we report all three regimes.

¹The SWE-bench Verified leaderboard archive contains traces and outcomes for over one hundred systems; we estimate it embodies \$50K–\$100K of agent compute contributed by submitters.

2 METHOD

3 TRACE REPRESENTATIONS ENCODE CONFOUNDERS

4 EFFICIENT EVALUATION

5 RELATED WORK

6 DISCUSSION

REFERENCES

- Brandon Duderstadt, Hayden S. Helm, and Carey E. Priebe. Comparing foundation models using data kernels. *arXiv preprint arXiv:2305.05126*, 2023.
- Hayden Helm, Brandon Duderstadt, Youngser Park, and Carey Priebe. Statistical inference on black-box generative models in the data kernel perspective space. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. arXiv:2410.01106.
- Hayden Helm, Johnson, and Carey Priebe. Query-efficient model evaluation using cached responses. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2605.07096.
- Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world github issues? In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.06770.
- Sayash Kapoor, Benedikt Stroebel, Zachary S. Siegel, Nitya Nadgir, and Arvind Narayanan. AI agents that matter. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2407.01502.
- Sayash Kapoor, Benedikt Stroebel, et al. Holistic agent leaderboard: The missing infrastructure for AI agent evaluation. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2510.11977.
- Alex Kipnis et al. metabench: A sparse benchmark to measure general ability in large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.12844.
- Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. tinyBenchmarks: Evaluating LLMs with fewer examples. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.14992.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019. arXiv:1908.10084.
- Mingjie Sun et al. Idiosyncrasies in large language models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.12150.
- Terminal-Bench Team. Terminal-bench: Benchmarking agents on hard, realistic tasks in command line interfaces. *arXiv preprint arXiv:2601.11868*, 2026.
- Sang Truong et al. Reliable and efficient amortized model-based evaluation. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2503.13335.
- Rajan Vivek, Kawin Ethayarajh, Diyi Yang, and Douwe Kiela. Anchor points: Benchmarking models with much fewer examples. In *Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2024. arXiv:2309.08638.
- Xingyao Wang et al. OpenHands: An open platform for AI software developers as generalist agents. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.16741.

John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2405.15793.

Richard Zhuang et al. EmbedLLM: Learning compact representations of large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.02223.